Coding & Solution

Date	06 July 2026
Team ID	TMID-SL-2026-01
Project Name	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

This section evaluates the quality and completeness of the implemented code and the overall solution.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/<your-username>/<your-repo-name>
Programming Language(s)	Python
Framework(s) Used	Flask, scikit-learn, XGBoost, imbalanced-learn, Pandas, NumPy
Key Features Implemented	EDA notebook; missing-value & categorical handling; SMOTE balancing; feature scaling; Decision Tree / Random
Pending / Incomplete Features	IBM Cloud deployment; authentication for credit officers vs. applicants; real-time credit bureau integration
Setup / Run Instructions	pip install -r requirements.txt → cd Training && python train_model.py → cd ../Flask && python app.py

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Partial
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments: The deployed model (XGBoost, tuned to n_estimators=150, max_depth=4, learning_rate=0.08) reaches 96.5% training accuracy and 81.0% test accuracy, selected for its balance of generalization and cross-validated performance among the four models compared.